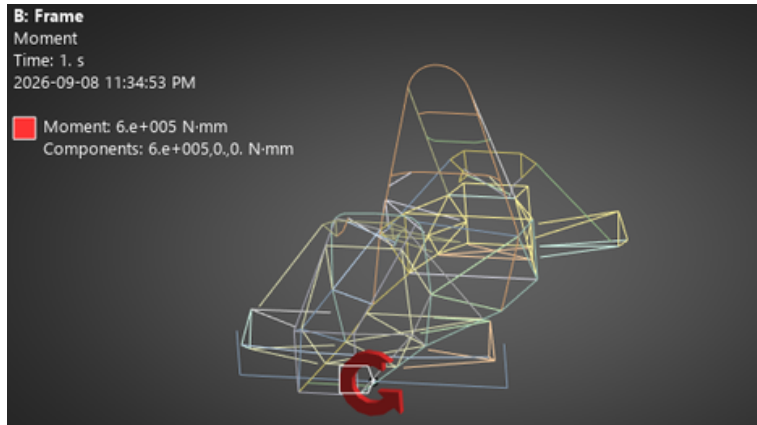


CHASSIS TORSIONAL STIFFNESS ANALYSIS

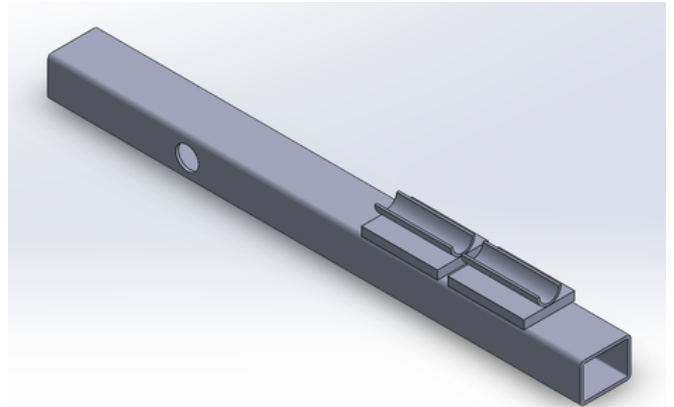
What?

- Evaluated torsional stiffness across four frame configurations; baseline, + AIP, + Accumulator, and combined, using ANSYS Finite Element Analysis (FEA) to guide stiffness-to-weight tradeoffs for the 2026 frame, and designed a physical fixture to enable future correlation against real-world testing

How?

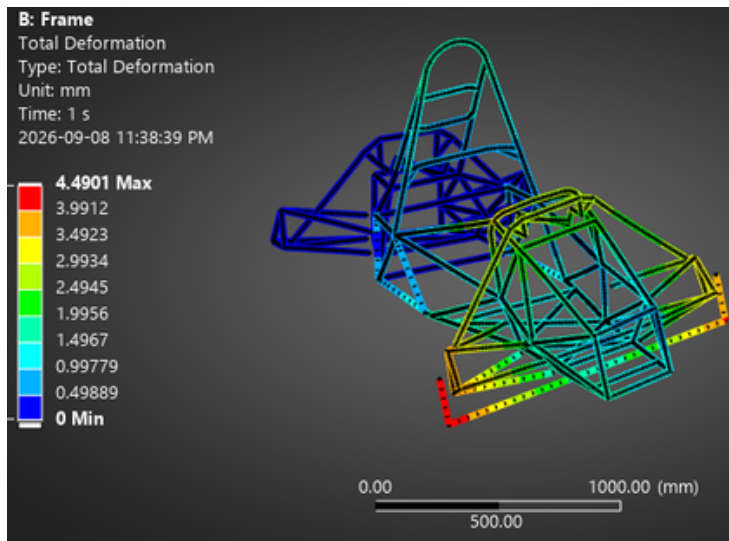


- Modeled the spaceframe with beam elements and spherical joints under a 600 N-m opposing moment

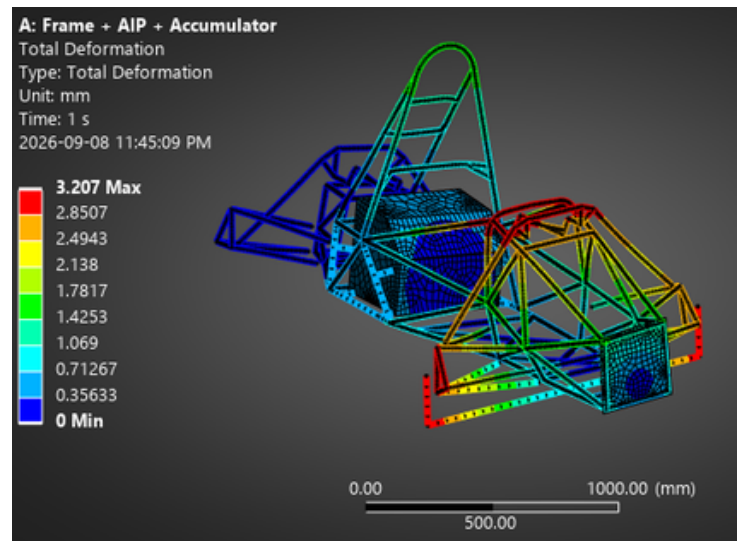


- Designed a rigid fixture to attach chassis to the rig, transferring pure torque without compliance errors

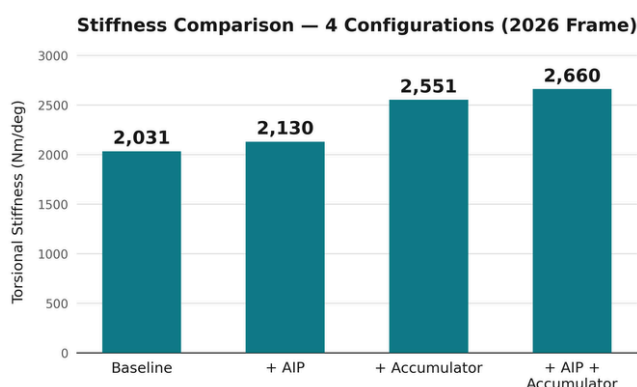
Results



Baseline - 4.49mm max deflection



Combined - 3.21mm max deflection



+25.6%

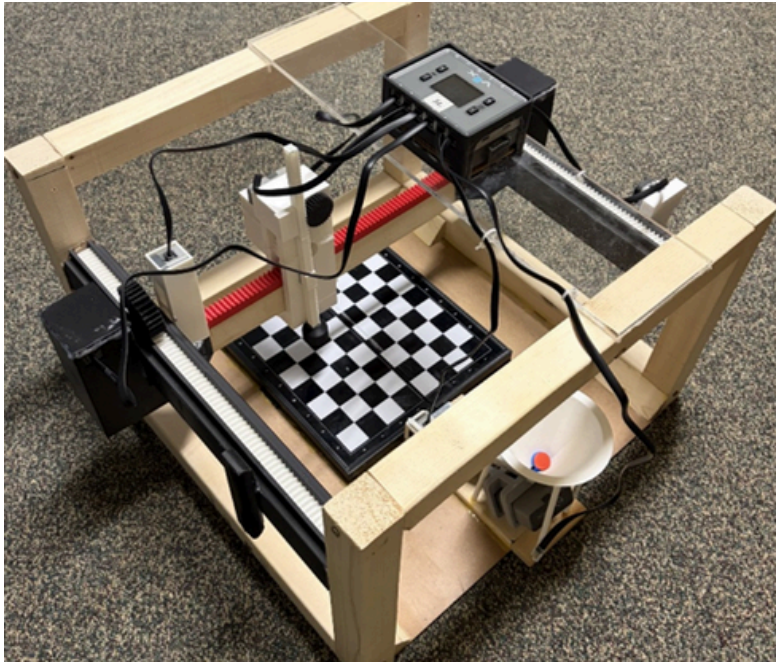
stiffness added by the accumulator alone

2660 Nm/deg

combined configuration, a 31% improvement over baseline

GANTRY-DRIVEN CHECKER BOT

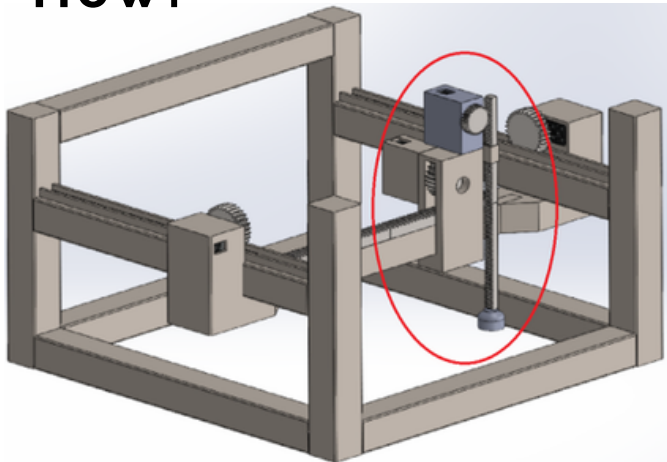
What?



Fully autonomous gantry robot placing all 24 checkers without human intervention. Owned complete mechanical CAD and physical assembly

- Gravity-fed hopper with rotating gate dispenser
- 3-axis rack and pinion gantry
- Colour detection for automated sorting
- Custom 90° bevel gear transmission

How?



Complete assembly designed in SolidWorks. Material selection by function:

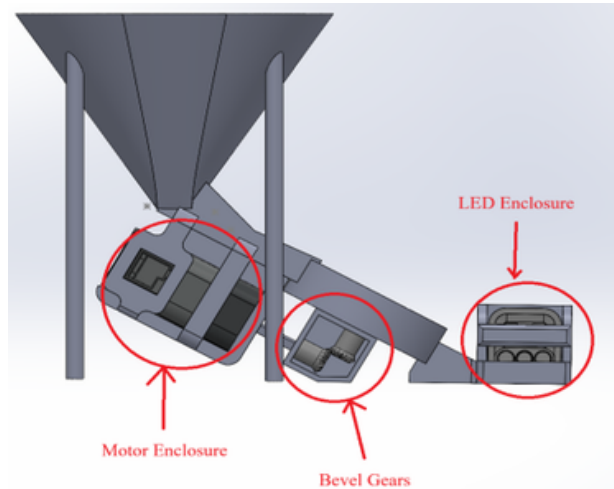
- PETG — structural racks
- Carbon fiber — high-stress pinions
- PLA — tolerance-critical sensor enclosures

Results

Specification	Target	Result	
Colour Detection	90%	94%	✓
Mechanical Reliability	90%	90%	✓
Total Mass	≤10 kg	7.01 kg	✓
Material Cost	<\$200	\$93.54	✓
Placement Accuracy	90%	75%	✗

Placement accuracy fell short due to gantry frame deflection.
Fix: metal guide rods along X and Y axes.

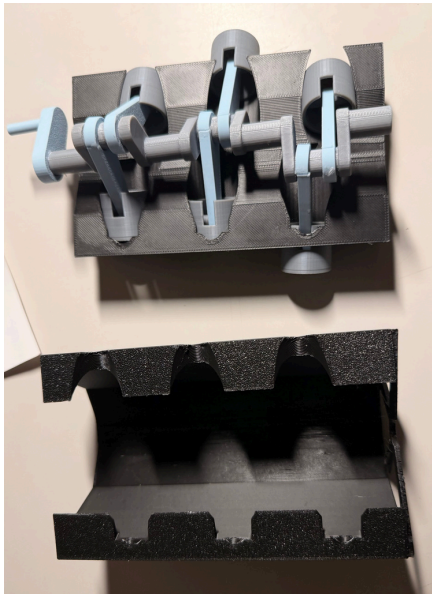
Details



Custom 90° bevel gear housing redirects motor torque to the rotating gate. PLA selected for tight gear mesh tolerances, custom printed rather than purchased.

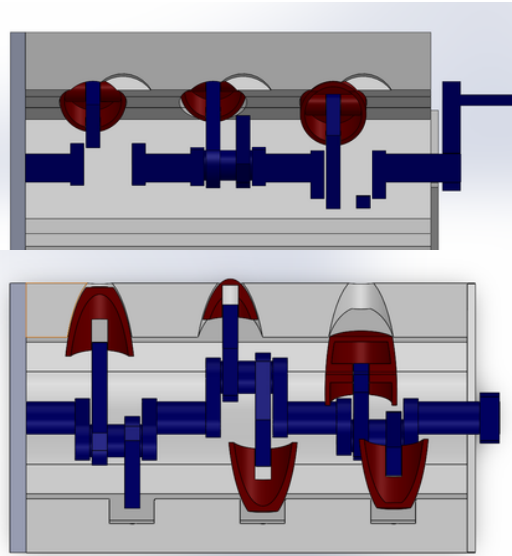
HAND-CRANK V6 ENGINE MODEL

What?



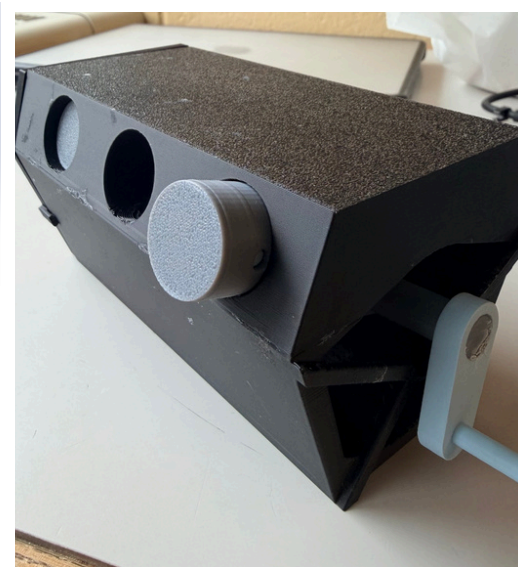
- Mechanically functional V6 assembly demonstrating accurate **4-stroke kinematics** and engine geometry

How?



- Designed in **SolidWorks** using **DFMA** principles for 3D printing
- Optimized rod-to-stroke ratio to minimize piston side loading
- Engineered **120° crankshaft phasing** for realistic firing order

Results



- Achieved **interference-free rotation** and accurate piston timing, validating geometric tolerance calculations

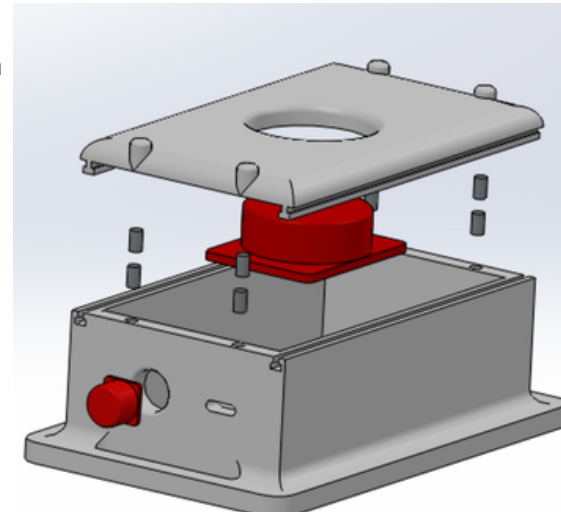
CARELINK (HEALTHTECH INNOVATION HACKATHON 2026)

What?

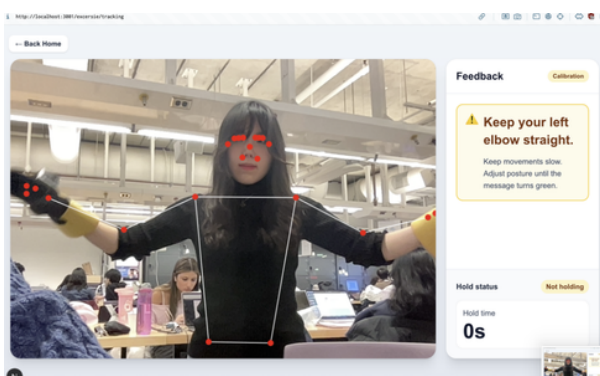


- Designed a sub-\$25 remote rehabilitation wearable for stroke survivors, utilizing hybrid sensor fusion (IMU + Computer Vision) to provide real-time, unsupervised exercise correction

How?



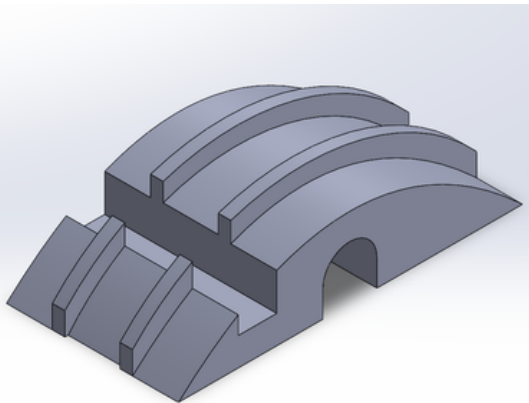
Results



- Delivered a fully functional hardware prototype within 48 hours that successfully filtered motion noise and maintained tracking accuracy during visual occlusion events
- Engineered a Class I medical-compliant housing for the ESP32/MPU6050 array in SolidWorks, prioritizing accessibility with a large physical calibration button and soft-touch EVA foam lining

BRAKE LINE ROUTING SYSTEM

What?



- Designed a custom **mounting block** to stabilize the brake lines. The geometry **constrains the zip tie** within a groove, preventing the line from sliding or vibrating

How?



- Prototyped** the mount on a steel plate first. This verified that the **zip-tie tension** provided enough grip to hold the tube rigid without slipping

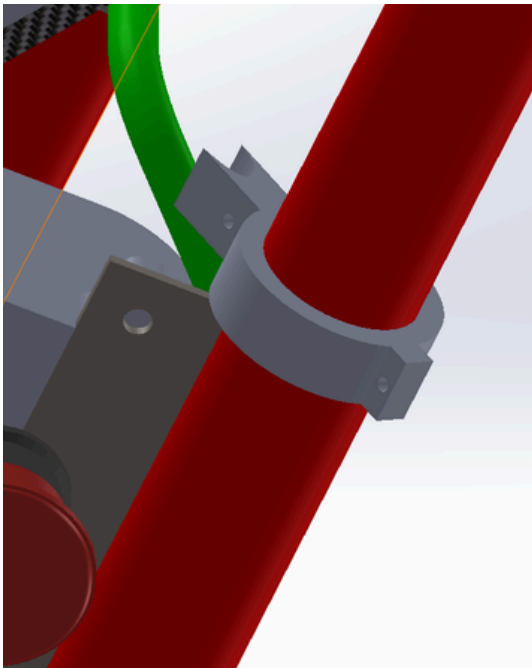
Results



- Finalized the design by drilling and installing it onto **carbon fiber**. This confirmed the mount fits correctly and holds securely to the actual **composite material** used on the car

WELDING FIXTURES

What?



How?



- Designed a **welding jig** to set the **maximum height** of the BRB Switch. Ensures precise vertical positioning relative to the chassis tube

- 3D printed** fixture creates a physical **hard stop** on the chassis tube. Constrains the steel tabs at the exact vertical limit, eliminating manual measuring errors

Results



- Achieved **<1mm vertical** accuracy, preventing rule failures during tech inspection. Reduced setup time to 5 mins with **zero post-weld adjustments**